- *Cloud Trace (https://cloud.google.com/trace)*—Google's offering for distributed tracing, which supports Go, Java, Node.js, and Python via OpenTelemetry.

Figure 5.4 provides an example output of Google Cloud Trace, a sophisticated distributed tracing backend. It shows the analytics of traces, specifically the latency distribution. Next, we will cover open source tracing solutions.

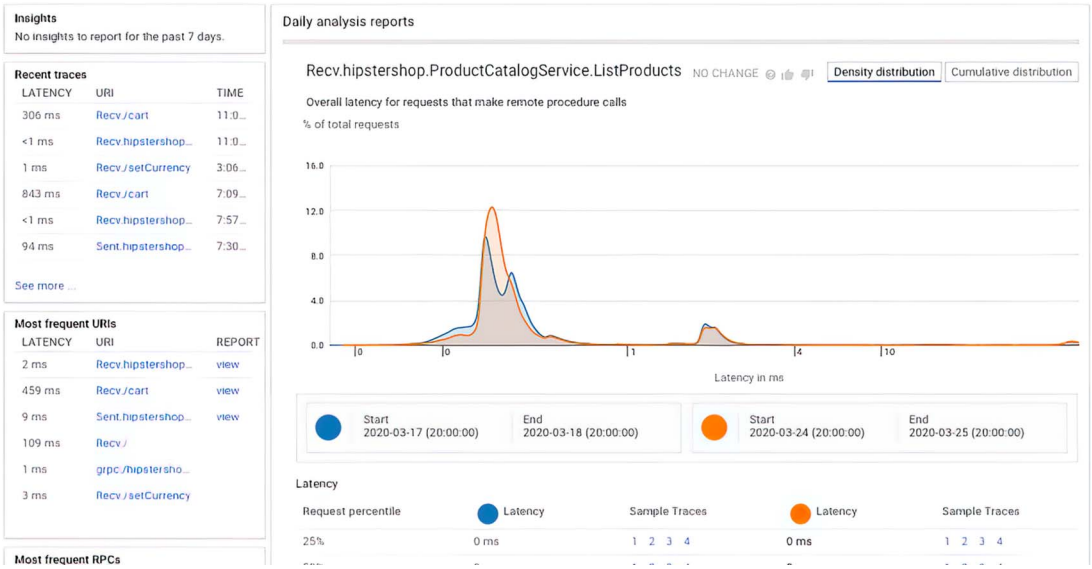


Figure 5.4 Viewing traces in Google Cloud Trace (Google Cloud docs; https://cloud.google.com/trace/docs/trace-overview)

5.4.2 Open source traces backends

Open source tracing solutions do not offer as many options as with metrics backends, but they still offer some choice:

- *CNCF Jaeger and Zipkin*—Both Jaeger (https://github.com/jaegertracing/jaeger) and Zipkin (https://github.com/openzipkin/zipkin) are distributed tracing systems hosted by CNCF. These days, Jaeger seems to be the more actively developed project, but Zipkin is still widely used and supported. I recommend you have a look at the respective communities (Slack and GitHub issues) and make up your own mind. We will cover both in chapter 6.
- *Elasticsearch and OpenSearch*—These options have expanded from logs to support handling traces. Both have excellent frontends (Kibana and Dashboards) that we will, again, cover in chapter 6.
- *Grafana Tempo (https://grafana.com/oss/tempo/)*—A distributed tracing backend available under the AGPLv3 license. It uses object storage (e.g., S3) in the